

D 120205

(Pages : 2)

Name.....

Reg. No.....

**SIXTH SEMESTER (CBCSS—UG) DEGREE EXAMINATION
MARCH 2025**

Physics/Applied Physics

PHY6B11/APH6B11—STATISTICAL PHYSICS, SOLID STATE PHYSICS, SPECTROSCOPY
AND PHOTONICS

(Admissions Year—2019 Onwards)

Time : Two Hours

Maximum : 60 Marks

*The symbols used in this question paper have their usual meanings.***Section A (Short Answer Type)***Answer **all** questions in two **or** three sentences, each correct answer carries a maximum of 2 marks.*

1. What are Bosons ? Give two examples.
2. Distinguish between Bose Einstein and Fermi Dirac statistics.
3. What is meant by unit cell ? Give an example.
4. Explain the terms crystal lattice and basis.
5. What is a symmetric top molecule ? Give an example.
6. What is zero point energy of a harmonic oscillator ?
7. What is isotopic substitution ?
8. List out some differences between laser light and ordinary light.
9. What is Morse curve ?
10. Explain the terms : (i) Population inversion ; and (ii) Metastable state.
11. Explain any *two* types of pumping mechanisms.
12. Optical pumping is not used in gas lasers. Why ?

(Ceiling = 20 marks)

Turn over

Section B (Paragraph / Problem Type)

Answer **all** questions in a paragraph of about **half a page to one page**, each correct answer carries a maximum of 5 marks.

13. Compare average velocity, root mean square velocity and most probable velocity.
14. Derive and explain Bragg's law.
15. The first line in the pure rotational spectrum of HCl appears at 21.18 cm^{-1} . Calculate bond length of the molecule. Given atomic masses of H and Cl is 1.008 and 35.45 amu respectively.
16. Explain the spectrum of a non - rigid rotator.
17. Calculate the bond force constant, k , for $^1\text{H}^{35}\text{Cl}$. The fundamental vibration frequency is $\overline{\nu}_e = 2990.9 \text{ cm}^{-1}$.
18. Obtain the equation for J_{max} , the rotational quantum number of the state to which the transitions give rise to spectral lines with maximum intensity.
19. Discuss the classical theory of Raman scattering.

(Ceiling = 30 marks)

Section C (Essay Type)

Essays - Answer in about **two pages**, any **one** question.

The question carries 10 marks.

20. With supporting diagrams, explain the working of a ruby laser.
21. Discuss the theory of a diatomic vibrating rotator.

(1 × 10 = 10 marks)